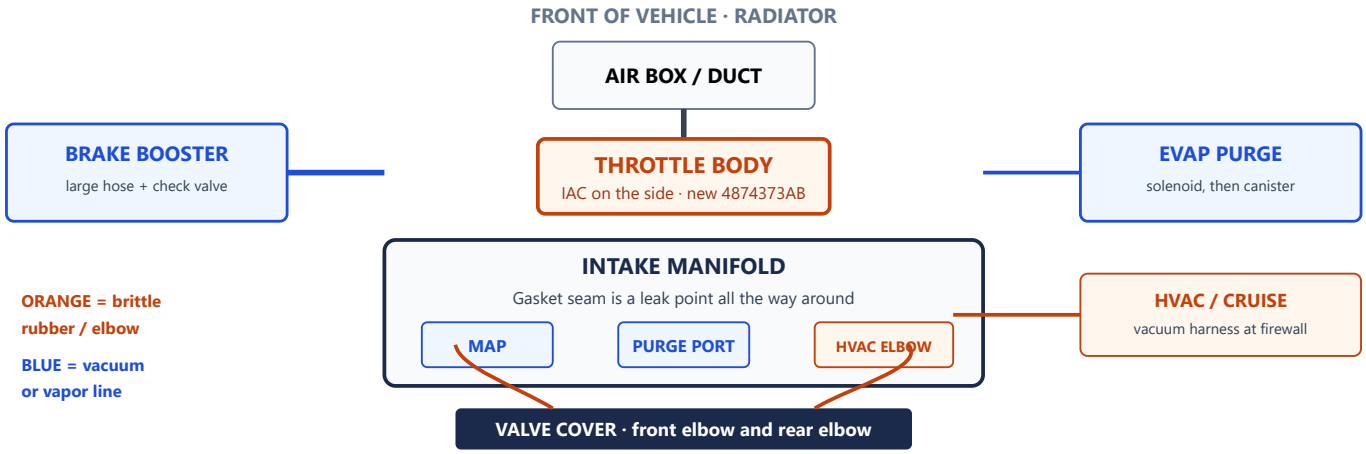
HOSE MAP · TYPICAL 4.0L ROUTING · CONFIRM EVERY LINE ON THIS TRUCK

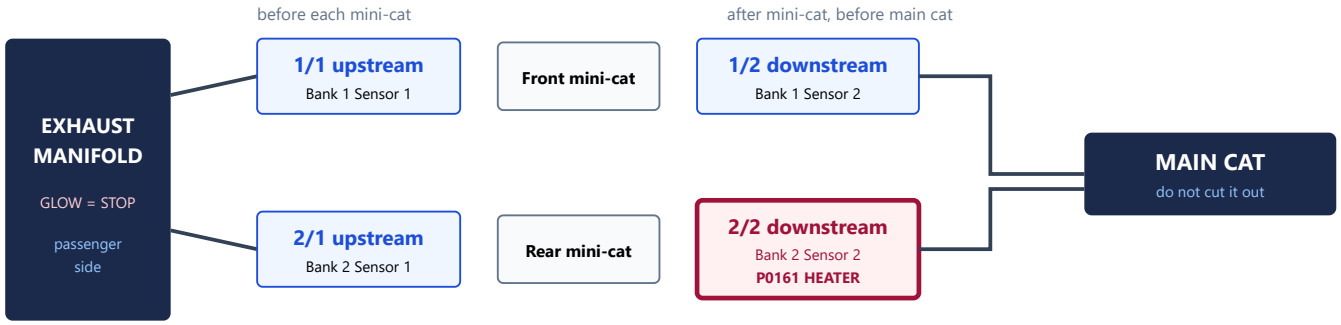
THIS IS A WORKING MAP, NOT A FACTORY VACUUM DIAGRAM · MATCH THE HOSE IN YOUR HAND



Line	Where it runs	Part to compare
CCV / PCV elbows	Valve cover, front and rear, into the intake or the air duct. The rear piece is short and often split. Oil vapor and vacuum both use this path.	Dorman 47128 / 4854265. Two short hoses, about 2.56 in. CATALOG LISTED Not the whole harness.
Brake booster	Large hose from the booster to a manifold port, with a check valve in the line. A split here raises idle and can make the pedal hard.	Match the hose and the valve. No number on this sheet.
EVAP purge	Purge solenoid on the engine, one hose to an intake port, one hose back toward the canister. A stuck-open valve acts like a vacuum leak.	Replace the cracked hose or the solenoid only after the test. Do not buy a canister for P0505.
HVAC / cruise	Small vacuum harness. The elbow and check valve are at the intake. A collapsed elbow kills HVAC and cruise. A cracked elbow can raise idle.	55115901AG only if the VIN is in the 31 Mar–6 Apr 2004 window. NARROW
IAC base	Not a hose. The O-ring under the motor is an air leak if it is missing, twisted, or flattened.	Comes with a new IAC. Do not push the pintle to seat it.

Ports on the drawing are grouped so the lines are readable. They are not a measured left/right layout. Fuel lines, heater hoses, and the power-steering hoses are not on this map. Do not pull those while chasing a vacuum leak.

PASSENGER-SIDE EXHAUST · FOUR SENSORS IF THIS IS THE CALIFORNIA PACKAGE



Oil that can reach this pipe

Trace fresh oil after a cold degrease. **LIKELY** sources, none confirmed until the wet trail starts there: valve-cover rail above the manifold, oil-filter adapter O-rings on this same side, oil-pressure sender, pan rail, rear main at the bellhousing. Fix the leak before a new O2 sensor goes in. No gasket numbers on this sheet. Order by VIN after the source is known.

Sensor purchase rule

Count sensors first. Two means federal. Four means the California layout drawn here. Buy one sensor, for the bung that failed the test, by VIN. Published 2004 4.0L candidates, position not mapped: 56028995AA, 56028994AA, 56028997AA, 05139021AA. **CANDIDATES** NTK 23142 and Mopar 56029001AB are **NOT CONFIRMED**.

Callout	What it is	Buy it when
1/1 and 2/1	Upstream sensors, before the mini-cats. These trim fuel. **LIKELY**	A signal or heater code names that exact sensor, and the circuit test fails the sensor.
1/2	Front downstream, after the front mini-cat. Watches that mini-cat.	Same rule. P0161 is not this sensor.
2/2	Rear downstream. Heater circuit is P0161.	Cold heater is open or shorted, or the sensor-side plug is destroyed, and oil is no longer hitting it.
Mini-cats and main cat	Three catalysts on the California package. **LIKELY**	Only after a temperature or pressure test shows a restriction. Not a first part. Do not delete them.
Manifold	Cast iron on the passenger side. A glow is a stop, not a parts order.	Replace only if it is cracked after it has been cool long enough to inspect. Guide 05.

Harness length and connector key are different at each bung. Do not swap sensors to see what happens. Clip every pigtail back so the wire cannot touch the pipe. Splice a broken wire with an adhesive-lined butt connector, several inches off the iron, staggered, then put the loom back.

PARTS LIST FOR THIS JOB · VERIFY EVERY NUMBER BY VIN

DO NOT HAND THIS PAGE OVER AS A BLIND ORDER · VIN AND THE PART IN YOUR HAND DECIDE

Item	Number	Use it for	Status
IAC motor, new	4874373AB	Pintle was moved, or a coil is open or shorted. Do not push the new pintle.	CATALOG LISTED
Short CCV hoses	Dorman 47128 Chrysler 4854265	The small cracked elbows at the valve cover / intake / MAP. About 2.56 in, two hoses.	CATALOG LISTED
O2, before catalyst	56028995AA 56028994AA	2004 4.0L published list. Which bank is which was not recoverable.	CANDIDATE
O2, after catalyst	56028997AA 05139021AA	Same list. One owner report tied 05139021AA to the rear downstream. Not proof.	CANDIDATE
HVAC vacuum line	55115901AG	Elbow and check valve only. Build dates 31 Mar–6 Apr 2004. Blocks HVAC and cruise. Does not fix a high idle by itself.	VIN WINDOW
Booster hose, purge hose, intake gasket, throttle-body gasket	Match by VIN	Buy the one the smoke test or the crack identifies. Not a set.	TEST REQUIRED
Valve cover, filter-adapter O-rings, sender, pan, rear main	Match by VIN after the fresh oil is traced	Oil-on-exhaust repair. Adapter and valve cover are the common ones. Rear main is the expensive miss.	TEST REQUIRED

Leave these numbers at home

NTK 23142 catalogs to other Jeeps, not this Grand Cherokee Bank 2 Sensor 2. **Mopar 56029001AB** had no fitment found. **53030006** is not a confirmed vacuum elbow for this truck. 53030008 is a PT Cruiser tee. NOT CONFIRMED

Wire repair, not a vehicle part

Broken heater or IAC wire: ratcheting crimper for heat-shrink butts, and adhesive-lined red butts, 22–16 AWG. Heat gun. Splice off the pipe. No solder sleeves, no T-taps, no tape-only joints, no grease in the O2 plug. A Weather-Pack assortment does not replace the O2 or IAC housing.

One hose is not the system

47128 does not include the booster hose, the purge line, or the HVAC harness. Replacing the short CCV pieces and leaving a split elbow at the booster still leaves the idle high. Smoke the manifold after the new pieces are on, then do the IAC and the relearn. Guides 03, 04, and 07.

Counter script. "2004 Grand Cherokee, 4.0L, this VIN. California package if I count four oxygen sensors. I need the Bank 2 Sensor 2, rear pipe, after the rear mini-cat" — or "the short CCV hoses, 4854265" — or "a new IAC, 4874373AB, and I will not let anyone push the pintle." If the counter's picture does not match the part in your hand, do not buy it.